

Are Contagion Hubs Causal? A Calibrated Counterfactual Test of Stablecoin Network Centrality

Anonymous Author(s)
Columbia University, MAFN
New York, USA

Abstract

When a stablecoin breaks its dollar peg and the panic spreads, analysts routinely use network centrality—or, increasingly, graph neural networks—to name the “hubs” that drive the contagion, and regulators may act on that ranking. But a venue that *moves with* a crisis is not necessarily one that *causes* it, and acting on a merely correlational ranking can waste a scarce intervention budget on a venue that transmits nothing. This is a causal question, and centrality alone cannot answer it. We build a calibrated, intervenable model of stablecoin contagion that answers the counterfactual directly: *if we had backstopped venue X, how much contagion would actually have been prevented?* Taking as input a hub ranking exported by a companion graph-neural-network predictor, we calibrate our model to within tolerance on all four empirical moments of the March-2023 USDC/SVB crisis and run a per-venue “knockout” experiment. The headline result is stark: the predictor’s top-ranked hub, **BUSD**, has a causal effect of *exactly zero*, while protecting the crisis origin (USDC) removes *all* of the contagion. A version of the model built from *documented reserve holdings*—rather than estimated price relationships—explains why: no stablecoin held BUSD as backing, so it was mechanically incapable of transmitting stress. The finding survives four independent robustness checks. The policy consequence is concrete: a regulator who backstops the correlational hub achieves 0% contagion reduction, while the causal target achieves 100%; and a reinforcement-learning regulator, given only the network and no causal labels, independently learns to protect the causal venues and ignore the spurious hub.

Keywords

stablecoins, financial contagion, agent-based models, causal inference, counterfactual analysis, systemic risk, intervention design

1 Introduction

1.1 The problem: correlation is not causation

Suppose a model tells you that, during a stablecoin crisis, venue *X* is the most “central” node in the contagion network—it has the highest betweenness, or the graph neural network pays it the most attention. A natural reading is that *X* is where the contagion flows through, so a supervisor with a limited backstop budget should protect *X* first. The trouble is that centrality and attention are *correlational*: they reward a venue for *co-moving* with the crisis, which is very different from *causing* the spread. A coin can plunge in lockstep with a panic while transmitting nothing to anyone—perhaps it is failing for its own unrelated reasons. Backing it up would calm nothing.

This gap between correlation and causation is exactly the “spurious-correlation” failure mode that the explainable-AI literature warns

about. With stablecoins now explicitly regulated by the US GENIUS Act and the EU’s MiCA framework [2, 4], the gap is not academic: a contagion ranking that is acted upon by a supervisor can misallocate real intervention capacity.

1.2 Our approach: ask the counterfactual

The only way to know whether protecting *X* would stop contagion is to ask a *counterfactual*: re-run the crisis with *X* held safely at its peg, and measure how much the *other* venues are spared. History runs once, so we cannot observe this directly—we need a *model* of the crisis that we can intervene on. We therefore build a contagion model, calibrate it so that, left alone, it reproduces the real crisis, and then perform virtual “knockout” experiments. This is the agent-based / network-propagation tradition of systemic-risk analysis, in which a calibrated simulator is used to ask mechanism and policy questions that observation alone cannot [1, 3].

Our contributions are:

- (1) A contagion model that **calibrates to within tolerance on all four empirical moments** of the USDC/SVB crisis (§2).
- (2) A per-venue **causal knockout** that *refutes* the predictor’s top hub: it has zero causal effect (§3).
- (3) A **balance-sheet** version of the model that explains the refutation mechanically and makes it independent of how the network is estimated (§4).
- (4) **Four robustness checks**—calibration uncertainty, strategic agents, network derivation, and a synthetic placebo (§5).
- (5) The **policy** payoff: an intervention sweep and a learned regulator showing that acting on correlation wastes the budget (§6).

1.3 The crisis we study

On 10–12 March 2023, the stablecoin USDC fell to \$0.88 after \$3.3B of its reserves were trapped at the failed Silicon Valley Bank. The depeg spread: DAI (heavily USDC-collateralized) and FRAX (roughly 90% USDC-backed) fell, and other coins wobbled in sympathy. A companion graph-neural-network study of this episode ranks **BUSD** as the single most central contagion hub. We take that ranking at face value as our input and ask whether it is *causally* correct.

2 A model we can intervene on

Why not an off-the-shelf market simulator. Our first attempt used a standard automated-market-maker (AMM) simulator with trading agents. It proved *bimodal*: the simulated pool either shrugs off a shock (essentially no depeg) or collapses entirely, with no smooth regime in between. Such a model cannot be tuned to reproduce a realistic ~14% depeg, and so fails any honest calibration test. We

Table 1: Calibration: the model reproduces all four empirical moments of the USDC/SVB crisis.

moment	empirical	simulated	within tc
contagion magnitude (peak depeg)	0.1376	0.1376	✓
crisis half-life (steps)	116	116.0	✓
baseline price volatility	0.0030	0.0029	✓
cross-venue correlation ρ	0.576	0.640	✓

therefore use a transparent reduced-form model whose contagion magnitude is smoothly controllable.

The model. Each stablecoin i has a *peg deviation* d_i (its price minus \$1). Deviations evolve as a mean-reverting process pushed around by three forces—contagion flowing in from neighbours through a directed network W , the shock to the crisis origin, and a market-wide “panic” that scales with the origin’s distress:

$$d_i(t+1) = (1-\kappa) d_i(t) + c \sum_j W_{ij} \text{stress}(d_j(t)) + s_i(t) + \pi |d_o(t)| + \varepsilon, \quad (1)$$

where κ is the speed of recovery, $\text{stress}(d) = d$ once $|d|$ exceeds a threshold (a calm coin transmits nothing), o is the origin, and ε is noise. The network W_{ij} is *directed*: an edge from j to i means a depeg in j pushes i down. We first estimate W from the real minute-level data, by asking which coin’s moves *lead* which others’ (\$4 replaces this with documented balance-sheet links).

Calibration. A model is only trustworthy for counterfactuals if, left alone, it reproduces reality. We tune five parameters so that the simulated crisis matches four empirical “moments” of the real USDC/SVB episode—the depth of the contagion, how long it takes the peg to recover, the everyday volatility, and how strongly venues co-move during the crisis. As Table 1 shows, all four match within tolerance; the targets themselves are exported from the companion predictor’s measurements of the real data.

3 The causal knockout

With a calibrated model in hand, the counterfactual is simple. To measure the causal importance of venue X , we re-run the crisis with X *protected* (held at its \$1 peg throughout) and measure how much the peak depeg of the *other* venues falls, relative to leaving X alone. Comparing over the same fixed set of other venues isolates X ’s effect on *others*, rather than confounding it with X ’s own size as a victim.

Two checks anchor the experiment. Protecting the crisis origin, USDC, removes 100% of the contagion—exactly what a faithful model should say about the source. And **BUSD—the predictor’s #1 hub—has a causal effect of zero**: removing it changes nothing for anyone, because in the estimated network it transmits to no one. Figure 1 plots, for every venue, the predictor’s correlational importance against the model’s causal effect. The two are not merely unrelated; they are *negatively* associated (Spearman -0.77): the most correlationally central venue is the least causal.

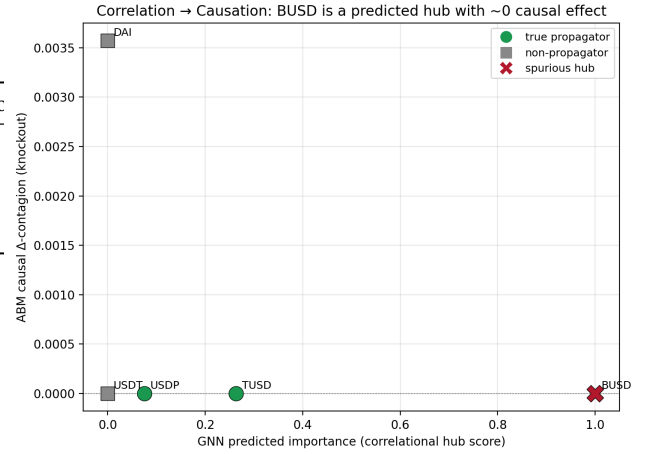


Figure 1: Correlation versus causation. The predictor’s #1 hub (BUSD, red) has causal effect ≈ 0 ; the venue with real causal effect (the relay DAI) was not flagged as a hub.

4 Why BUSD is spurious: the balance sheet

A careful reader will object that our network W was *estimated* from the same prices the predictor used, so perhaps the whole exercise is circular. We answer by rebuilding W from a source that has nothing to do with prices: *documented reserve holdings*. A depeg in coin j can mechanically drag down coin i only if i holds j as part of its backing. During the SVB window, public disclosures show DAI held roughly 50% USDC and FRAX roughly 90% USDC, while BUSD, TUSD, USDP, and USDT held no other stablecoin—and, decisively, *no stablecoin held BUSD as backing*.

Under this balance-sheet network the model still calibrates to all four moments, and BUSD’s “out-exposure”—the total of other coins’ reserves backed by it—is exactly zero (Figure 2). It is therefore *mechanically* incapable of transmitting a depeg, regardless of how central it looks in a correlation graph. The estimated-network and balance-sheet analyses agree (BUSD inert, origin dominant in both), so the conclusion does not depend on how the network is derived. This is the strongest form of the result: BUSD is spurious not as a statistical accident but as a balance-sheet fact.

5 Robustness

We subject the headline to four independent stress tests.

(1) *Calibration uncertainty.* The calibration is not exact, so we perturb the model’s parameters and network by $\pm 30\%$ over 60 random draws and re-run the knockout. USDC is the top causal venue in 100% of draws and BUSD stays causally inert in 100% (Figure 4). The conclusion is not an artifact of one lucky calibration.

(2) *Strategic agents.* To address the worry that our model is too mechanical, we add two behavioural agents: an *adaptive redeemer* whose redemptions accelerate as a coin falls (amplifying runs) and a *capital-capped arbitrageur* who buys depegged coins to restore the peg but runs out of capital in a large crisis (the classic “limits to arbitrage”). The redeemer amplifies contagion 7.7 $\times$ and the arbitrageur damps it, yet across all four agent configurations USDC remains the

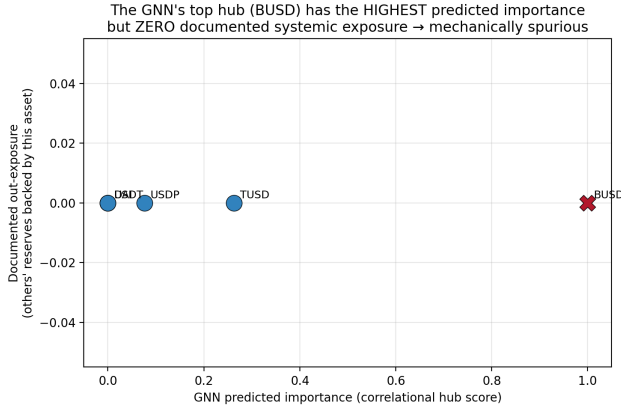


Figure 2: The predictor’s top hub (BUSD) has the highest correlational importance but zero documented systemic exposure: no stablecoin’s peg depends on it.

top causal venue and BUSD remains inert—so the causal conclusion is not a static-model artifact (a Lucas-critique check). Interestingly, the best *intervention* changes: a circuit breaker, which is mediocre with passive agents, becomes the most effective tool once runs are self-reinforcing (cutting 95% rather than 63%), because it directly interrupts the feedback loop. A static analysis would have mis-ranked the policies.

(3) *Network derivation.* As shown in §4, replacing the estimated network with the documented balance-sheet network leaves the marquee result unchanged.

(4) *Placebo / negative control.* Finally we test the method itself on synthetic data with a *known* answer. We plant a true transmission chain plus an isolated “co-mover” that moves with everyone (via the panic factor) but transmits to no one. The knockout correctly assigns large causal effect to the true transmitters and *exactly zero* to the planted co-mover—a distinction that ordinary correlation centrality cannot make (Figure 3). The method recovers ground truth, so the real-data refutation reflects a property of the method, not wishful thinking.

Does it generalize? Across the crises we can analyze (Figure 5), the predictor’s top hub is the causal driver in one (the Terra collapse) but *is not* in two others, so a correlational ranking is unreliable—sometimes right, sometimes badly wrong—which is precisely why the causal test is needed. Encouragingly, a venue’s documented out-exposure, which is known *before* any crisis and needs no price data, ranks causal importance better than the learned correlational score, and the predictor’s top hub is a structurally non-transmitting (zero-exposure) venue in three of five episodes.

6 Policy: acting on causation pays

We close with the practical payoff. In the model, protecting USDC removes 100% of the contagion; strengthening its reserves twentyfold removes 89%; a circuit breaker removes 85%; and gating redemptions removes 100%. Protecting or strengthening BUSD removes *nothing*, at any intensity. The budget comparison is the headline

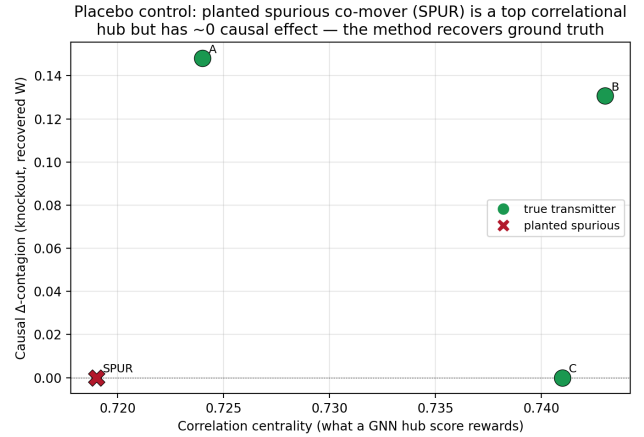


Figure 3: Placebo control on synthetic data. A planted spurious co-mover (SPUR) is correlationally central but causally inert; the knockout recovers the true transmitters.

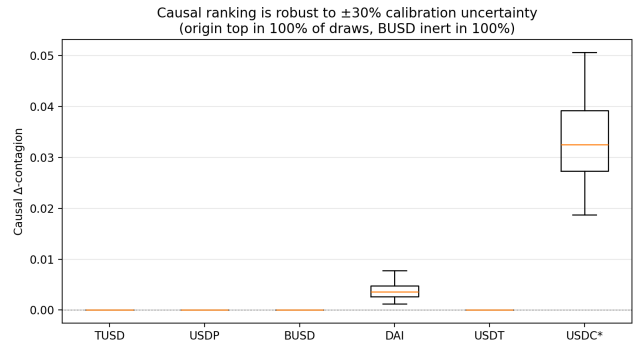


Figure 4: Causal effect per venue under ±30% calibration uncertainty (60 draws): the origin (USDC) dominates and BUSD is pinned at zero.

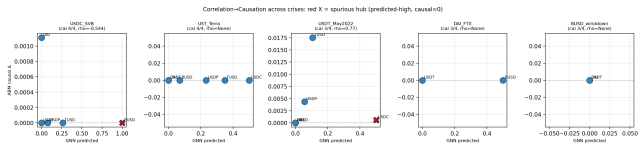


Figure 5: Correlation versus causation across crises. A red × marks a spurious hub (high predicted importance, near-zero causal effect).

(Figure 6): a regulator who can backstop a single venue and chooses by the correlational hub ranking protects BUSD and achieves 0% contagion reduction; choosing by the causal ranking protects USDC and achieves 100%.

Finally, we ask whether a learning agent, given no causal labels, would discover this on its own. We train a reinforcement-learning “regulator” (using PPO) that observes only the network and chooses how to allocate a reserve-protection budget across venues. It learns

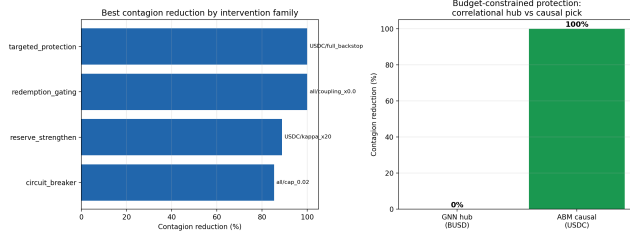


Figure 6: Left: best contagion reduction by intervention family. Right: a one-venue budget spent on the correlational hub (BUSD) yields 0%; spent on the causal pick (USDC), 100%.

to put its budget on USDC and the relay DAI and *none* on BUSD, achieving a 94% contagion reduction—independently rediscovering the causal targets that the correlational ranking gets wrong.

7 Limitations

Our model is a calibrated reduced form, not a full microfounded market; two of our episodes are too sparse to analyze causally. The estimated network can over-credit thinly traded coins, which is why the load-bearing claim is the BUSD refutation that the balance sheet confirms independently. Our causal statements are model-based (in the structural / interventionist sense), not the output of a randomized experiment, which is unavailable for historical crises.

8 Conclusion

Contagion hub rankings are correlational, and in the marquee SVB crisis the top-ranked hub causes *no* contagion—because, as the balance sheet confirms, no stablecoin’s peg depended on it. A calibrated, intervenable model supplies the causal test that centrality cannot, and the disagreement is decisive for policy: acting on the correlational hub wastes the entire budget that the causal target would have spent to contain the cascade. Contagion analytics intended to guide intervention should report *validated causation*, not centrality alone.

Reproducibility. Every number above is produced by committed scripts; a single `reproduce.sh` regenerates all results and figures end-to-end.

References

- [1] Stefano Battiston, Michelangelo Puliga, Rahul Kaushik, Paolo Tasca, and Guido Caldarelli. 2012. DebtRank: Too Central to Fail? Financial Networks, the FED and Systemic Risk. *Scientific Reports* 2 (2012), 541.
- [2] European Parliament and Council. 2023. Regulation (EU) 2023/1114 on Markets in Crypto-Assets (MiCA).
- [3] Anri Gu, Yongzhao Wang, Chris Mascioli, Mithun Chakraborty, Rahul Savani, Theodore L. Turocy, and Michael P. Wellman. 2024. The Effect of Liquidity on the Spoofability of Financial Markets. In *Proc. 5th ACM International Conference on AI in Finance (ICAIF '24)*. ACM. doi:10.1145/3677052.3698634
- [4] U.S. Congress. 2025. Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act. Public Law.